



Goal

Provide deeper scientific insights into conductivity, dielectric behavior, and stability relationships.

KEY INSIGHTS

Stability vs. Elasticity (looking for "Indestructible" materials)

Bottom-Right. These are materials that are thermodynamically "happy" (0 on the hull) but are physically as stiff as a rock:

- X-Axis (Stability): 0 is perfectly stable.
- Y-Axis (Elasticity/Stiffness): Higher is stiffer.

Band Gap vs. Dielectric Constant

Looking for "Capacitor" materials—materials that can store an electric charge without letting it "leak" through

- For electronics, we want the High Dielectric Constant (stores energy) but with a Moderate Band Gap (so it doesn't short-circuit)

Ferroelectric Candidate Discovery ()

Ferroelectric materials are used in how high-speed computer memory (FeRAM) and specialized sensors

- Every material in this list that have high stability (low hull energy) — those are the ones that can be manufactured easily

Refractive Index Analysis

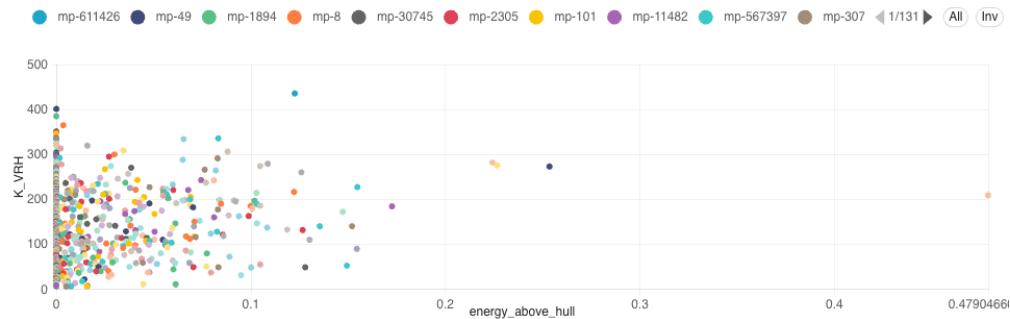
It is Optical Properties (how light moves through the material):

- material on the far right (high refractive index) = candidate for advanced camera lenses, fiber optics, or anti-reflective coatings

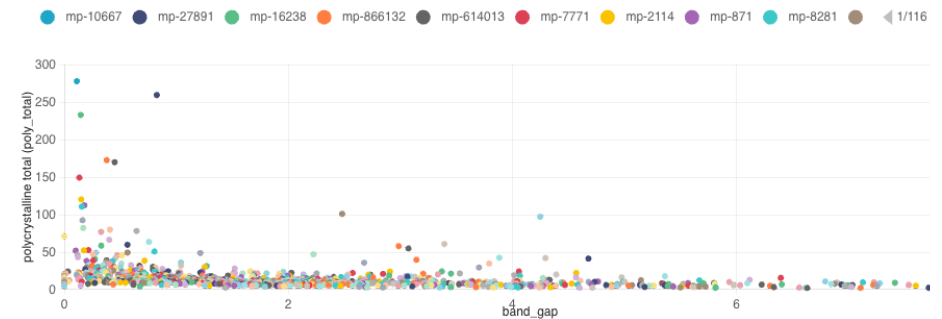
Key Business Questions

- What properties correlate with conductivity?
- Which materials are stable and mechanically robust?
- Which materials may be promising ferroelectrics?

Stability vs Elasticity



Band Gap vs Dielectric Constant



Ferroelectric Candidate Discovery

standard_formula	energy_above_hull	band_gap	poly_total
AgAsF6	0	1.680599999999992	4.29
AgPF6	0	3.2361	4.38
Al2CdS4	0	2.726299999999997	8.23
Al2CdSe4	0	2.004699999999999	9.31
Al2CdTe4	0	1.6245	10.72
Al2HgSe4	0	1.425399999999991	12.09
Al2HgTe4	0	1.1108	13.59
Al2ZnSe4	0	2.1529	9.18
Al2ZnTe4	0	1.625099999999999	10.85
AlAgO2	0	1.409099999999991	10.73



Refractive Index Analysis

